# The Anthropogenic Reservoir

**Why the Most Profitable Feedstock on Earth Is the Material We've Already Processed**

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## Executive Brief

The global industrial economy is built on an extractive model: dig raw materials from the earth, process them at enormous energy cost into useful forms, deploy them once or twice, and discard them. This model treats geological extraction as the default supply pathway and end-of-life material as waste.

This framing is a category error with trillion-dollar consequences.

The material we have already extracted, refined, and processed — metals in scrap yards, carbon chains in plastic waste, rare earths in electronics, minerals in demolition rubble — constitutes an enormous, energy-rich reservoir of industrial feedstock. We call it the **anthropogenic reservoir**.

The core thesis of this paper is simple:

> **Accessing processed materials from the anthropogenic reservoir requires a fraction of the energy needed to extract and refine equivalent materials from geological sources. Lower energy input means lower cost. Lower cost means higher margin. The anthropogenic reservoir is not a sustainability story. It is a profitability story.**

Key findings:

- **Recycled aluminum** requires approximately 5% of the energy of primary production from bauxite ore.
- **Scrap steel** processed in electric arc furnaces uses roughly 25–30% of the energy of blast furnace reduction from iron ore.
- **Plastic pyrolysis** to recover usable hydrocarbon chains bypasses the entire extraction-refining-cracking sequence that crude oil must undergo, at substantially lower total energy cost.
- **Copper recovery** from scrap avoids the mining, crushing, concentrating, and smelting chain — estimated at 10–15% of primary production energy.
- These are not marginal savings. They represent **order-of-magnitude reductions** in the single largest variable cost in materials production: energy.

The barrier to capturing this value is not technological. Pyrolysis, electric arc furnaces, and metals reprocessing are mature, proven technologies. The barrier is **perspective** — the institutional, regulatory, and conceptual frameworks that separate "mining," "petrochemicals," and "waste management" into distinct industries operating on what is, at the molecular and atomic level, a single unified material field.

The separation of "waste," "resource," and "feedstock" is not a physical distinction. It is a domain declaration error — a mispartitioning of a single material field into institutional categories that obscure the underlying energy gradient. When the domain is corrected to include all processed material as available state, that gradient becomes economically actionable.

This paper makes the case that any firm continuing to rely exclusively on geological feedstock in material classes where secondary processing is available is not merely missing an opportunity — it is operating at a structural cost disadvantage that widens with every year of declining extractive returns.

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## 1. Introduction: The Extractive Assumption

Modern industrial supply chains are organized around a foundational assumption: useful materials come from the earth. Ore bodies supply metals. Oil reservoirs supply carbon chains. Mineral deposits supply construction inputs. The entire upstream infrastructure of the global economy — exploration, drilling, mining, smelting, refining — exists to convert geological raw materials into industrially useful forms.

This assumption was reasonable in the 19th century, when the total stock of processed materials in human hands was negligible compared to geological reserves. It is no longer reasonable today.

Over 150 years of industrial activity, humanity has extracted, refined, and deployed an enormous quantity of material. The global stock of steel in use exceeds 35 billion tonnes. Cumulative plastic production has surpassed 10 billion tonnes, with a substantial fraction persisting in landfills and the environment. Billions of tonnes of aluminum, copper, zinc, and other metals exist in manufactured form across buildings, vehicles, infrastructure, and electronics.

This material has already absorbed the energy cost of extraction and refinement. The chemical bonds in ore have already been broken. The carbon chains have already been assembled. The atoms are already in their industrially useful states.

When these materials reach end-of-life, the conventional model classifies them as waste — a disposal problem to be managed. This paper argues that classification is the single most

expensive mistake in the modern industrial economy.

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## 2. The Energy Gradient: Why Anthropogenic Feedstock Is Cheaper

### 2.1 The Fundamental Principle

Every material processing step — mining, smelting, refining, cracking, polymerizing — requires energy input to overcome chemical or physical barriers. The total energy embedded in converting a raw geological resource into a finished industrial material is the **primary processing energy cost**.

When that material reaches end-of-life, most of that energy investment is retained in the material's chemical and physical state. Metallic bonds are still metallic. Carbon-carbon bonds are still intact. Crystal structures are still formed.

Returning end-of-life material to a useful industrial state requires overcoming only the barriers specific to reprocessing — typically remelting, purification, or controlled thermal decomposition. These barriers are categorically smaller than the barriers overcome during primary processing.

This creates a persistent **energy gradient** favoring anthropogenic feedstock over geological feedstock for equivalent output materials.

### 2.2 Carbon Chains: Plastic Waste vs. Crude Oil

The conventional pathway from crude oil to plastic polymer involves:

1. **Extraction** — drilling, pumping, artificial lift
2. **Transportation** — pipeline or tanker to refinery
3. **Distillation** — atmospheric and vacuum separation into fractions
4. **Cracking** — steam cracking of naphtha at 800°C+ to produce ethylene and propylene
5. **Polymerization** — catalytic assembly of monomers into polymer chains under heat and pressure

Each step carries energy costs, thermal losses, and infrastructure overhead.

Plastic waste to usable hydrocarbon chains via pyrolysis involves:

1. **Collection and sorting** — the primary logistical cost
2. **Thermal decomposition** — controlled heating at 400–600°C in the absence of oxygen

The pyrolysis step produces a hydrocarbon mixture — pyrolysis oil — with chain lengths comparable to diesel and naphtha fractions. This oil can feed directly into existing petrochemical infrastructure for further processing, or be used as fuel.

The total energy input for pyrolysis-based recovery is substantially less than the full crude-to-polymer chain, because the most energy-intensive steps — geological extraction and the initial cracking of large molecules into reactive monomers — are bypassed entirely. The carbon-hydrogen bonds already exist. The molecular architecture is already partially organized.

The primary constraint on plastic pyrolysis at scale is feedstock consistency and sorting efficiency, not thermodynamic feasibility. Mixed and contaminated plastic streams require preprocessing that adds cost and complexity. But this is a logistics problem with known engineering solutions — not a fundamental barrier to the energy economics described above.

If the end product is new polymer, the pyrolysis oil requires repolymerization — but the feedstock still arrives at the polymerization step at lower total energy cost than virgin naphtha.

If the end product is fuel-length carbon chains, the energy savings are even more pronounced: one thermal step replaces an entire extraction-to-refining sequence.

### 2.3 Metals: Scrap vs. Ore

The energy advantage of scrap metal over virgin ore is well-documented but systematically undervalued in strategic planning.

**Aluminum** is the most dramatic case. Primary aluminum production from bauxite requires:

1. Mining and crushing bauxite ore
2. Bayer process digestion to extract alumina ($Al_2O_3$)
3. Hall-Héroult electrolytic reduction at ~950°C with massive electrical input (~15 kWh per kg of aluminum)

Secondary aluminum production from scrap requires:

1. Collection and sorting
2. Remelting at ~660°C (aluminum's melting point)

Energy requirement: approximately 0.75 kWh per kg — roughly **5% of primary production**.

**Steel** follows a similar pattern. Blast furnace reduction of iron ore to pig iron, followed by basic oxygen furnace steelmaking, requires approximately 20–25 GJ per tonne. Electric arc

furnace steelmaking from scrap requires approximately 6–8 GJ per tonne — roughly **25–35% of primary production**.

**Copper** primary production involves mining, crushing, flotation concentration, smelting at 1200°C+, and electrolytic refining. Secondary copper from clean scrap requires remelting and modest refining. Energy savings are estimated at **85–90%** relative to primary production.

These are not theoretical projections. They are operational realities already demonstrated at industrial scale. Electric arc furnace steelmaking already accounts for approximately 30% of global steel production, driven entirely by the cost advantage of scrap feedstock.

### 2.4 The Unified Gradient

Across material categories, the pattern is consistent:

| Material | Primary Energy (Geological) | Secondary Energy (Anthropogenic) | Approximate Savings |
|---|---|---|---|
| Aluminum | ~175 GJ/tonne | ~8 GJ/tonne | ~95% |
| Steel | ~20–25 GJ/tonne | ~6–8 GJ/tonne | ~65–75% |
| Copper | ~35–50 GJ/tonne | ~4–7 GJ/tonne | ~85–90% |
| Plastic (to fuel-grade) | Full extraction + refining chain | Pyrolysis at 400–600°C | ~50–70% (estimated) |

The gradient is not marginal. It is structural. And it applies to every material class where the anthropogenic reservoir holds significant inventory.

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## 3. The Inventory: What We've Already Built

### 3.1 Scale of the Anthropogenic Reservoir

The anthropogenic reservoir is not a hypothetical future resource. It is a present physical reality of extraordinary scale.

**Plastics:** Cumulative global plastic production through 2025 exceeds 10 billion tonnes. Of this, an estimated 4–5 billion tonnes persists in landfills, and a significant additional quantity exists in the environment. Current annual plastic waste generation exceeds 350 million tonnes. Even with low collection rates, the accessible inventory is measured in billions of tonnes of reduced carbon chains.

**Steel:** The global in-use stock of steel is estimated at over 35 billion tonnes, with annual end-of-life flows exceeding 500 million tonnes. This stock is continually renewed as buildings, vehicles, and infrastructure reach end-of-life.

**Aluminum:** Global in-use aluminum stock exceeds 1 billion tonnes, with annual scrap generation exceeding 30 million tonnes.

**Copper:** Global in-use copper stock exceeds 500 million tonnes, with annual scrap availability exceeding 10 million tonnes.

**Rare earths and specialty metals:** While smaller in absolute tonnage, the concentration of rare earth elements in electronics waste frequently exceeds their concentration in mined ore by factors of 10–100x. A tonne of circuit boards contains more gold than a tonne of gold ore, more copper than a tonne of copper ore, and meaningful quantities of palladium, platinum, and rare earth elements.

### 3.2 The Inventory Is Growing

Unlike geological reserves, which are finite and depleting, the anthropogenic reservoir grows with every product manufactured and every structure built. Every new car, building, electronic device, and plastic container adds to the future feedstock inventory. The anthropogenic reservoir is the only major material source on Earth whose total stock increases over time.

This creates a compounding structural advantage. As primary geological reserves become more energy-intensive to access — deeper wells, lower-grade ores, more remote deposits — the anthropogenic reservoir simultaneously grows in volume and maintains its low reprocessing energy cost.

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## 4. Why the Market Hasn't Corrected: The Perspective Barrier

If the energy economics are this favorable, why isn't the market already dominated by secondary material flows? The answer is not technological. It is structural and perceptual.

### 4.1 Institutional Fragmentation

The extractive industries, manufacturing industries, and waste management industries evolved as separate institutional domains with separate:

- Regulatory frameworks
- Professional cultures and identities
- Capital structures and investment models

- Lobbying and policy influence channels
- Academic disciplines and research funding streams

A mining company sees ore. A waste hauler sees tonnage. A petrochemical company sees feedstock specifications. No institutional actor is structured to see the unified material field — the total inventory of atoms and molecules in industrially useful states, regardless of which institutional category currently claims them.

### 4.2 Accounting Conventions

Current accounting frameworks classify extracted raw materials as assets and end-of-life materials as liabilities (disposal costs). This classification is a projection-layer artifact — it reflects institutional convention, not physical reality. It is, in structural terms, the same class of error that causes any system to misprice assets: the domain over which value is declared does not match the domain over which value exists. Correcting the domain declaration — recognizing end-of-life material as feedstock in a different state rather than waste requiring disposal — is not an ideological position. It is an accounting correction with immediate economic consequences.

A landfill containing 10 million tonnes of mixed plastic is, at the molecular level, a hydrocarbon reservoir with known composition and zero exploration risk. Under current accounting conventions, it appears on no balance sheet as a resource. The entity responsible for it books disposal costs, not feedstock value.

Reversing this accounting treatment — recognizing anthropogenic material stocks as assets with quantifiable energy-cost advantages over geological alternatives — would immediately revalue billions of dollars in currently written-off material.

### 4.3 The Identity Problem

Perhaps most significantly, the transition from extractive to reprocessing models challenges deep professional and cultural identities. Oil companies are in the oil business. Mining companies mine. These identities are reinforced by decades of institutional history, workforce training, investor expectations, and regulatory classification.

Reframing these enterprises as material-state management companies — entities that specialize in moving atoms and molecules between useful configurations at minimum energy cost — is conceptually straightforward but culturally radical.

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## 5. The Opportunity: Structural Cost Advantage

### 5.1 The Arbitrage

The current market prices materials based primarily on the cost of primary production from geological sources, adjusted for supply and demand dynamics. Secondary materials trade at discounts that reflect perceived quality risk, sorting costs, and the stigma of "recycled" classification.

This pricing structure undervalues secondary materials relative to their thermodynamic reality. The entity that builds processing capacity against anthropogenic feedstock — at its true, lower energy cost — holds a structural margin advantage against every competitor paying the geological extraction premium.

This is not a bet on regulation, carbon taxes, or consumer preference for sustainability. It is a straightforward energy-cost arbitrage available under current market conditions.

### 5.2 Declining Extractive Returns

Geological extraction is subject to declining returns. The easiest, richest, shallowest deposits were developed first. Each successive generation of extraction targets lower-grade ores, deeper reservoirs, more remote locations, and more technically demanding formations. The energy cost of primary production trends upward over time.

The anthropogenic reservoir exhibits the opposite dynamic. As collection infrastructure improves, sorting technology advances, and processing techniques mature, the energy cost of secondary production trends downward over time — while the total inventory grows.

These are converging curves. The crossover point — where secondary production is unambiguously cheaper than primary production across all major material categories — is approaching rapidly. For some materials (aluminum, steel), it has already arrived.

### 5.3 Risk Profile

Geological extraction carries exploration risk, geopolitical risk, permitting risk, and depletion risk. The anthropogenic reservoir carries none of these. The material is already located (in cities and landfills), already within existing jurisdictions, already characterized in composition, and growing rather than depleting.

The risk-adjusted return profile of anthropogenic feedstock processing is structurally superior to geological extraction for equivalent output materials.

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## 6. Implementation Pathways

### 6.1 Near-Term: Expand Proven Processes

Electric arc furnace steelmaking, aluminum remelting, and copper scrap processing are already industrial-scale operations. The near-term opportunity is simply to expand these operations and redirect capital from primary extraction to secondary processing capacity.

### 6.2 Medium-Term: Plastic Pyrolysis at Scale

Plastic pyrolysis technology is commercially proven but not yet deployed at the scale the feedstock inventory warrants. Building large-scale pyrolysis capacity — treating plastic waste as a hydrocarbon reservoir rather than a disposal problem — represents the single largest near-term opportunity in carbon chain economics.

### 6.3 Long-Term: Urban Mining for Specialty Materials

Electronics waste contains concentrations of precious and rare earth metals that exceed natural ore grades by orders of magnitude. As automated sorting and hydrometallurgical recovery processes mature, urban mining of electronics waste will become the primary supply pathway for critical technology metals.

### 6.4 Infrastructure Reframing

Existing waste collection and processing infrastructure — landfills, transfer stations, materials recovery facilities — should be reconceived as the upstream supply chain for secondary materials production. This reframing does not require new physical infrastructure in most cases. It requires new operational models, different capital allocation, and revised regulatory classification.

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## 7. Conclusion

The extractive model served humanity well during the build-out phase of industrial civilization. It is now the more expensive option for the majority of bulk materials.

The anthropogenic reservoir — the total stock of processed materials in human hands and human waste streams — is the largest, lowest-energy, lowest-risk feedstock source available to the global economy. It is growing while geological reserves deplete. Its energy economics are superior by factors of 2x to 20x depending on the material class.

The companies, investors, and policymakers who recognize this first will capture a structural cost advantage that compounds over time. Those who continue to treat geological extraction as the default supply model will find themselves paying an increasing

energy premium for materials that their competitors source from the anthropogenic reservoir at a fraction of the cost.

The material is already here. The energy math is already settled. The only remaining question is not whether the industrial economy will pivot from geological extraction to anthropogenic reprocessing — it is who will capture the margin advantage of doing so first, and who will pay the compounding cost of being last.

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**Metatron Dynamics** *Relational analysis applied to structural reality.*

*The analytical framework underlying this paper — including the domain declaration methodology and relational operator structure used to identify the unified material field — is developed and maintained by Metatron Dynamics. For technical documentation, see relationalrelativity.dev.*